

# Lecture 20

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## *Determinant III & Linear Transformation I*

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# Outline and Review

5 mins

# Roadmap of Semester

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**Segment 1** Solving linear systems (Lec 3-15);

**Segment 2** **Lec 16-22:** Three relatively independent parts:

- **Orthogonality**. Least squares; orthogonal matrix (Lect 16,17,18)
- **Determinant**. [Important tool!] Lec 18,19, 20
- **Linear transformation**. Lec 20-22 [Advanced math perspective of matrix]

They are not directly related to solving system/problem,  
but are **fundamental and useful tools**.

**Segment 3** **Lec 23-28: Eigenvalues and singular values.**

# Today's Lecture: Learning Goals

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**Main topic:** Determinant (III) and Linear transformation (I)

1. Advanced Properties of Determinant
2. Definition of Linear Transformation
3. Matrix of Linear Transformation

Strang's book: Sec 8.1, 8.2

**After the lecture, you should be able to**

1. Tell determinant of product and inverse
2. Describe one application of linear transformation
3. Verify whether a mapping is linear transformation
- 4 Compute the matrix of linear transformation

# Definition of $\det(A)$

Let  $A \in \mathbb{R}^{n \times n}$  be a square matrix.

Denote by  $M_{ij} \in \mathbb{R}^{(n-1) \times (n-1)}$  a matrix formed by deleting the  $i$ -th row and  $j$ -th column of  $A$ , called the **minor (余子式)** of  $a_{ij}$ .

## Definition 19.1 (Determinant)

For a scalar  $\alpha \in \mathbb{R}$ , define  $\det(\alpha) = \alpha$ .

For any  $A \in \mathbb{R}^{n \times n}$  with  $n \geq 2$  define  $|A| = \det(A)$  as

$$\begin{aligned}\det(A) &= \sum_{j=1}^n (-1)^{1+j} \det(M_{1j}) a_{1j} \\ &= a_{11} \det(M_{11}) - a_{12} \det(M_{12}) + a_{13} \det(M_{13}) - \cdots + (-1)^{1+n} \det(M_{1n}) a_{1n}.\end{aligned}$$

This is a **recursive definition**! [递归方式的定义]

$M_{1j}$  is an  $(n-1) \times (n-1)$  matrix;

its determinant is a weighted sum of determinant of  $(n-2) \times (n-2)$  matrices;

Continue the process, until we reach  $1 \times 1$  determinants (which are just the entries themselves).

# Basic Properties of Determinant

**Property P19.0 [transpose]**  $\det(A^T) = \det(A)$

**Property P19.1 [row/column-linear]**

The determinant is a linear function of each row and each column separately.

**Property P19.2 [reduction]** .

If an  $n \times n$  matrix  $A = \begin{bmatrix} h & \mathbf{0}_{1 \times (n-1)} \\ \mathbf{0}_{(n-1) \times 1} & B \end{bmatrix}$ , then  $\det(A) = h \cdot \det(B)$  .

**Property P19.3 [interchange]**  $\det(E_{R_i \leftrightarrow R_j} A) = -\det(A)$

**Corollary 19.1 [Addition]**  $\det(E_{\beta R_j + R_i} A) = \det(A)$

“Add a Scaled Row to Another”  
does not change the determinant.

**Corollary 19.2 [Scaling]** .  $\det(E_{\alpha R_i} A) = \alpha \cdot \det(A)$

**Remark:** since  $\det(A^T) = \det(A)$ ,

all properties on rows **apply to columns**

(so the corollaries applies to column operations as well).

# Part I Advanced Properties of `det()`

10 mins

# Desired Properties

## Property P20.1 [triangular matrix]

If  $A \in \mathbb{R}^{n \times n}$  is a triangular matrix with diagonal entries  $a_{11}, \dots, a_{nn}$

$$\det(A) = a_{11} \dots a_{nn}$$

$$\begin{bmatrix} a_{11} & \times & \times \\ 0 & a_{22} & \times \\ 0 & 0 & a_{33} \end{bmatrix} = a_{11} \begin{bmatrix} a_{22} & \times \\ 0 & a_{33} \end{bmatrix} = \dots$$

$$\det(A+B) \neq \det(A) + \det(B)$$

## Property P20.2 [product]

For any two matrices  $A, B \in \mathbb{R}^{n \times n}$ ,  $\det(AB) = \det(A)\det(B)$

## Property P20.3 [inverse]

For any invertible  $A \in \mathbb{R}^{n \times n}$ ,  $\det(A^{-1}) = \frac{1}{\det(A)}$

$$\begin{aligned} \det(A) \det(A^{-1}) &\stackrel{P20.2}{=} \det(A \cdot A^{-1}) \\ &= \det(I) \stackrel{P20.1}{=} 1 \end{aligned}$$

# Determinant and Invertibility

**Proposition** For any  $A \in \mathbb{R}^{n \times n}$ ,  $\det(A) \neq 0$  iff  $A$  is invertible.

**Geometry:**  $\det(A)$  = volume of polytope

volume = 0  $\iff$  polytope collapses to a lower-dimension polytope [via polytope]  
 $\iff$  vectors are linearly dependent [via vectors]  
 $\iff$   $A^{-1}$  does not exist [via matrix A]

**Algebra:** Use GJ-E.

GJ-E transforms  $A$  to REF  $U$ , which is an upper triangular matrix.

elementary row operation multiplies the determinant by 1, -1, or nonzero scalar,

so  $|A| = (-1)^c \alpha |U|$  for some  $\alpha \neq 0$ , integer  $c$ .

So  $|A| \neq 0$  iff  $|U| \neq 0$   $U_{ii}$ 's are diagonal entries of  $U$

$\iff U_{11} U_{22} \cdots U_{nn} \neq 0 \iff U_{ii} \neq 0, \forall i$  [Property P20.1]

$\iff$  there are  $n$  pivots  $\iff A$  is invertible.

# Equivalent Conditions for Invertibility ++

## Theorem (Equivalent Conditions for Invertibility)

Let  $A \in \mathbb{R}^{n \times n}$

The following statements are equivalent:

1.  $A$  is **invertible**
2. The linear system  $A\mathbf{x} = \mathbf{0}$  has a unique solution  $\mathbf{x} = \mathbf{0}$
3.  $A$  is a product of elementary matrices
4.  $A$  has  $n$  pivots; or equivalently:  $\text{rank}(A) = n$
5. The columns of  $A$  span  $\mathbb{R}^n$
6. The columns of  $A$  are linearly independent
7. The columns of  $A$  form a basis of  $\mathbb{R}^n$
8.  $\dim(C(A)) = n$
9.  $\dim(N(A)) = 0$  or  $N(A) = \{\mathbf{0}\}$
10.  $A\mathbf{x}=\mathbf{b}$  is solvable for any  $\mathbf{b}$
11.  **$\det(A) \neq 0$**

# Example: Product of Determinant

## Property P20.2 [product]

For any two matrices  $A, B \in \mathbb{R}^{n \times n}$ ,  $\det(AB) = \det(A)\det(B)$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \\ 3 \cdot 2 + 4 \cdot 1 & 3 \cdot 0 + 4 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 10 & 4 \end{pmatrix}$$

$$\det(AB) = 4 \cdot 4 - 2 \cdot 10 = -4$$

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(B) = 2 \cdot 1 - 0 \cdot 1 = 2$$

$$\det(A) \det(B) = (-2)(2) = -4$$

## Reading material:

### Proof Sketch of $\det(AB) = \det(A) \det(B)$

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#### Two cases

**Case I:**  $A$  is not invertible.

Then  $AB$  is not invertible (exercise).

Then  $\det(A)\det(B) = 0$ ; and  $\det(AB) = 0$ . The relation holds.

**Case II:**  $A$  is invertible.

Write  $A = E_m \dots E_1$  where each  $E_i$  is an elementary matrix.

**Lemma:** If  $E$  is elementary matrix, then  $\det(EK) = \det(E)\det(K)$  for any square matrix  $K$ .

$$\begin{aligned}\det(AB) &= \det(E_m \cdots E_2 E_1 B) \\ &= \det(E_m) \cdot \det(E_{m-1} \cdots E_2 E_1 B) \\ &\quad \vdots \\ &= \det(E_m) \cdots \det(E_2) \cdot \det(E_1) \cdot \det(B) \\ &= \det(E_m \cdots E_2 E_1) \cdot \det(B) \\ &= \det(A) \cdot \det(B).\end{aligned}$$

# Properties of Determinants: Overview

## Special matrices

Diagonal matrix

Triangular matrix

Block matrix

Elementary matrix

## Elementary operations

Interchange two rows or two columns

Adding a scaled row or column

Scaling a row or column

## Matrix operations

$$A + B$$

$$A \cdot B$$

$$A^{-1}$$

$$A^T$$

Single-row-sum

single-column-sum

$$\det(A+B) \neq \det(A) + \det(B)$$

$$\det(AB) =$$

$$\det(A^{-1}) =$$

$$\det(A^T) =$$

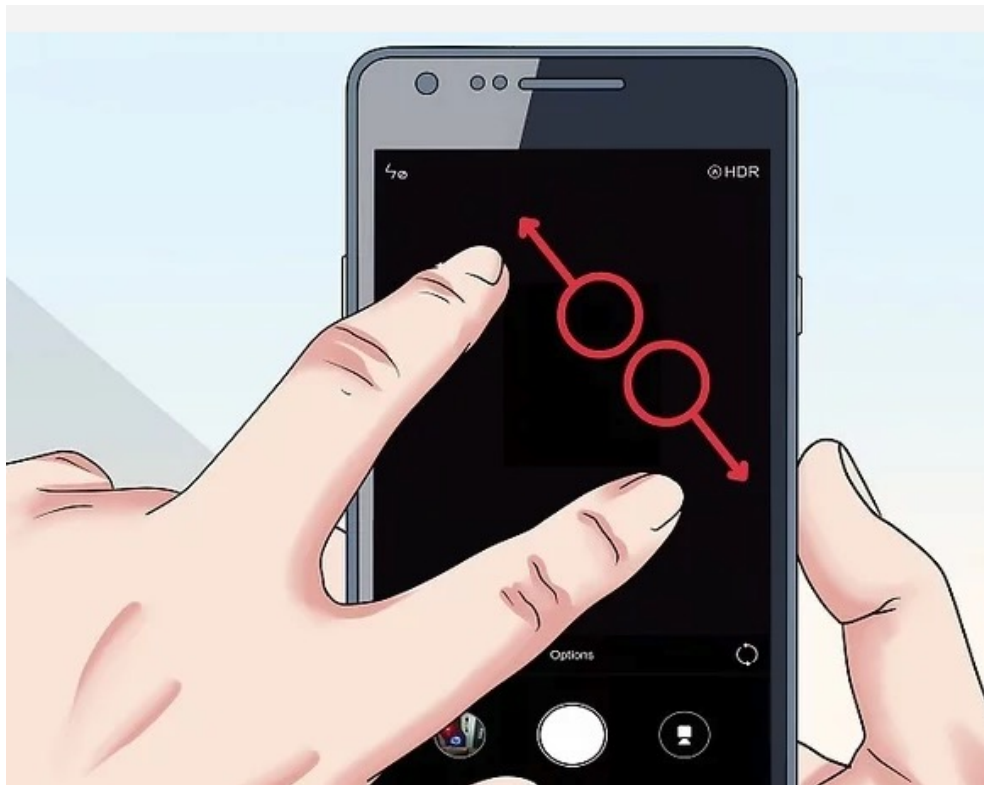
# Part II Motivation of Linear Transformation

5 mins

# Zoom Photos

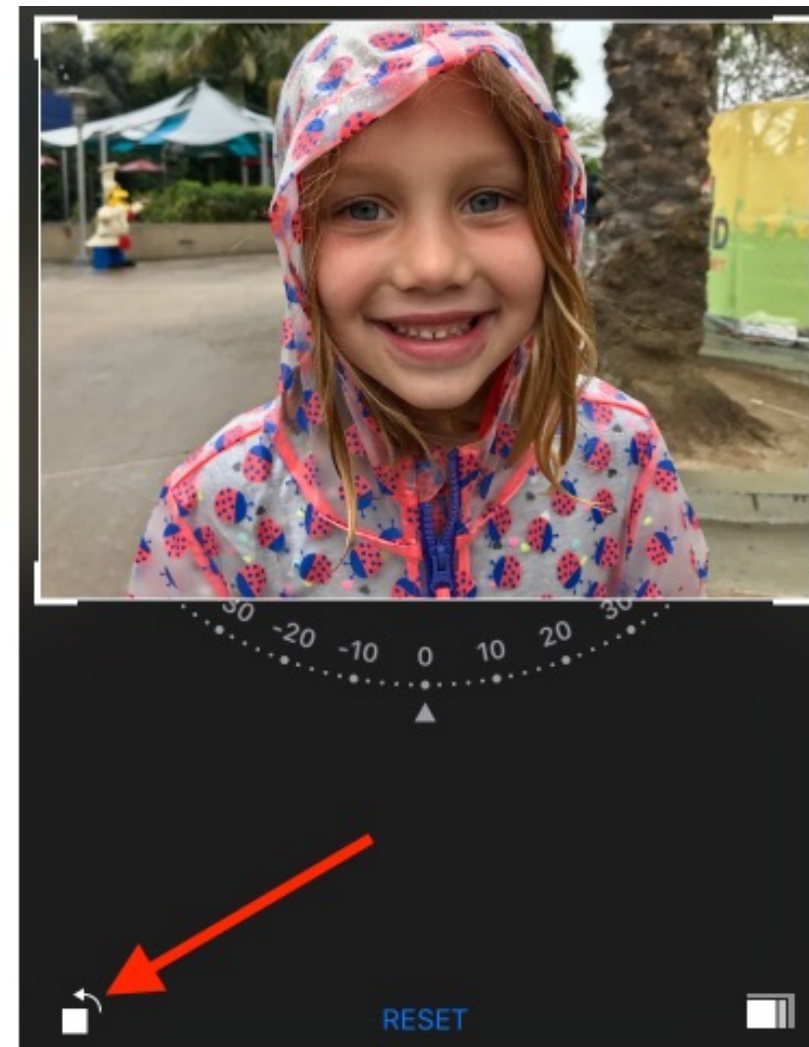
ANDROID » ANDROID APPLICATIONS

## How to Zoom with the Camera on Android



# Rotate Photos

## How to rotate photos on iPhone



# Motivating Question

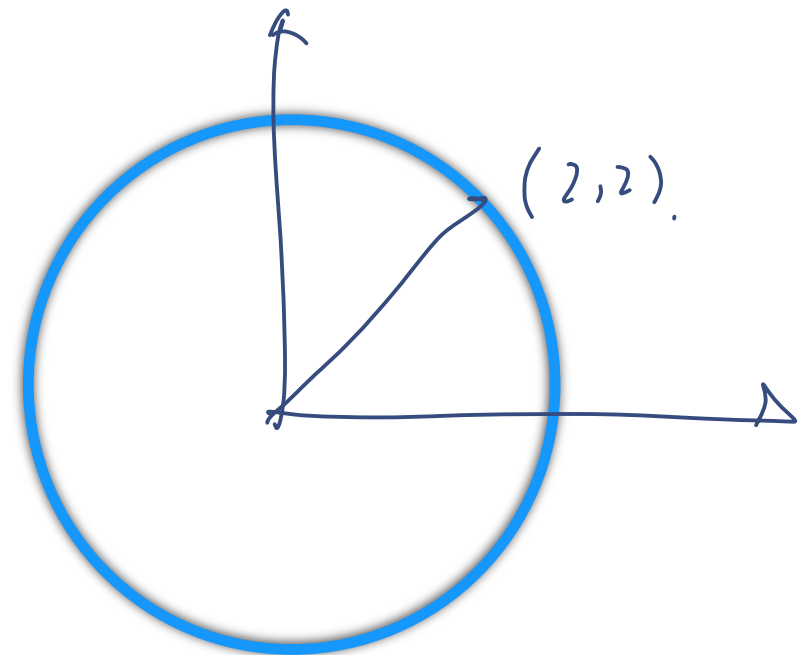
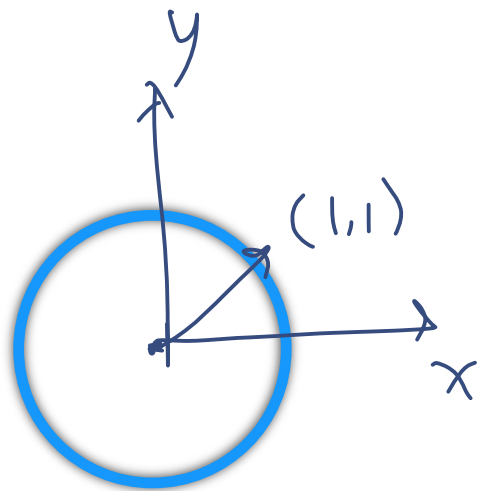
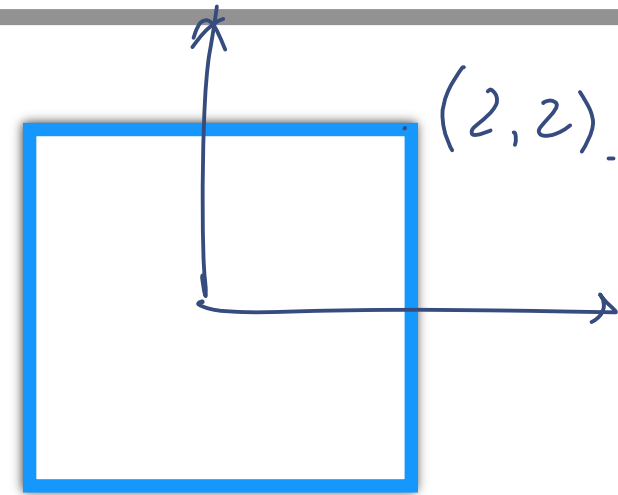
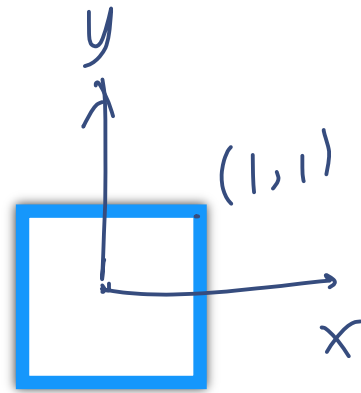
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Yes, I know you can do it on the phones.

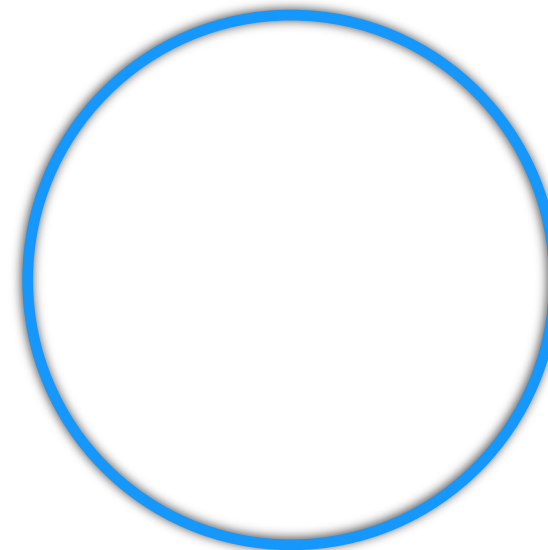
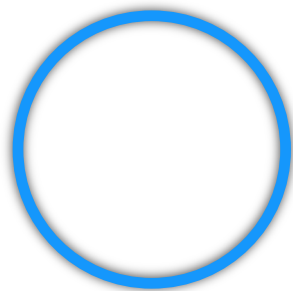
But...

How do the phones accomplish the job?

# Zoom Shapes



# Zoom Shapes

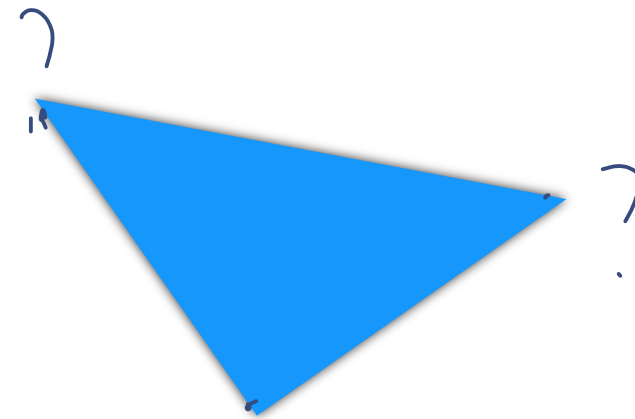
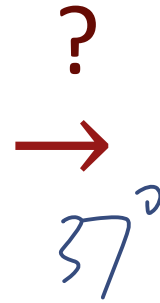
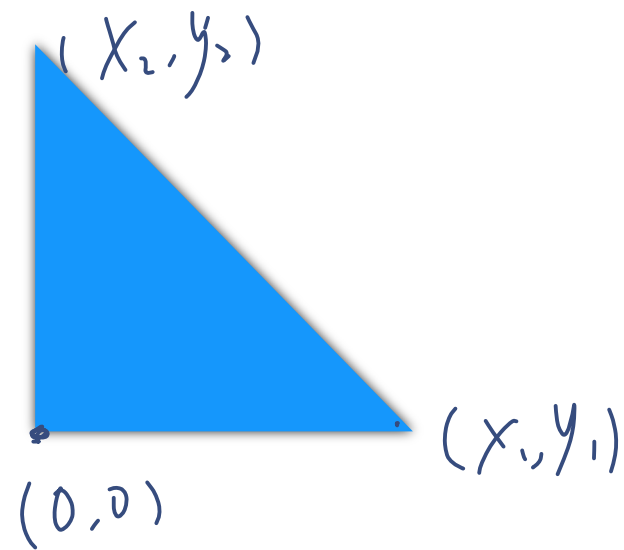


$(x,y)$



$(2x,2y)$

# Rotating Shapes



# Rotating Shapes



$(x, y)$



Why true?  
How to come up?

$$(x \cos 35^\circ - y \sin 35^\circ, \\ x \sin 35^\circ + y \cos 35^\circ)$$

# Part III Definition of Linear Transformation

10 mins

# Recall: Function

A major part of math courses in middle & high school is about functions.

The definition of function in typical textbooks:

## Function (an earlier definition):

**English:** Given two non-empty **sets of numbers**,  $X$  and  $Y$ , if there exists a **certain definite correspondence relation  $f$** , such that **for every number in set  $X$ , there is a uniquely determined number  $f(X)$**  in set  $Y$  corresponding to it, then this relation  $f: X \rightarrow Y$  is called a function.

**Chinese:** 设两个非空数集 $X, Y$ , 如果按照某种确定的对应关系 $f$ , 使得对于 $X$ 中的任意一个数, 在 $Y$ 中都有唯一确定的一个数 $f(x)$ 与之对应, 那么就称 $f: X \rightarrow Y$ 是一个函数。

It emphasizes two key ideas:

1. Each input ( $x$ ) is allowed

2. Each input has a **unique** output ( $y$ )

Find  $y$  that  $y^2 = x$ .  $x \mapsto y$

not a function

# Mapping

## Definition (Mapping)

Let  $V$  and  $W$  be two **sets**. If for any  $v \in V$ , there is a **unique**  $w = T(v) \in W$ , then we say  $T$  is a **mapping** from  $V$  to  $W$ .

$V$  is called the **domain** (定义域) of the mapping.

$W$  is called the **codomain** (陪域) of the mapping.

We denote:  $T: V \rightarrow W$   $\longrightarrow$   
 $v \mapsto T(v)$   $\longmapsto$

## Special case:

Suppose  $f_i$  is a function from  $\mathbb{R}^n$  to  $\mathbb{R}$ ,  $i = 1, \dots, m$ ,  
then  $f: x \mapsto (\underline{f_1(x)}, \dots, \underline{f_m(x)})$  is a **mapping** from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ .

# Examples of Mapping

## Example 1: Matrix multiplying vector.

- **Domain ( $V$ ):**  $\mathbb{R}^3$  (3D space)
- **Codomain ( $W$ ):**  $\mathbb{R}^2$  (2D plane)
- **Rule:**  $T(\mathbf{v}) = A\mathbf{v}$ , where  $A$  is a  $2 \times 3$  matrix.

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

## Example 2: Neural Network Layer.

In machine learning, a single “layer” of a neural net is a mapping that transforms a high-dimensional input.

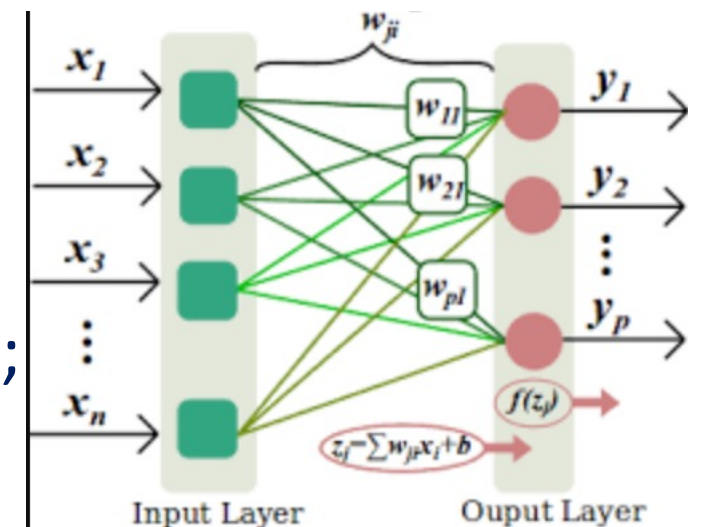
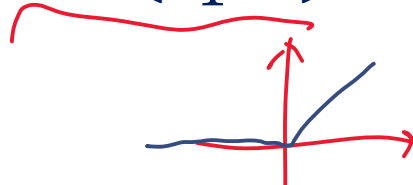
**Domain ( $V$ ):**  $\mathbb{R}^n$  (Input features)

**Codomain ( $W$ ):**  $\mathbb{R}^m$  (Hidden units)

$$T(x) = \sigma(Wx + b)$$

where  $\sigma(\cdot)$  is a “neuron” function, e.g., ReLU function;

$$\sigma(z_1, \dots, z_n) = (\max(z_1, 0), \dots, \max(z_n, 0)).$$



# Linear Transformation

## Definition 20.1 (linear transformation):

If a mapping  $f$  from a linear space  $V$  to a linear space  $W$  satisfies

$$f(\underbrace{\alpha \mathbf{x} + \beta \mathbf{y}}_{\text{LC of } \mathbf{x} \& \mathbf{y}}) = \underbrace{\alpha f(\mathbf{x}) + \beta f(\mathbf{y})}_{\text{LC of } f(\mathbf{x}) \& f(\mathbf{y})}, \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in V,$$

Then we say  $f$  is a **linear transformation** from  $V$  to  $W$ .

It satisfies: transformed LC of vectors = LC of transformed vectors.

## Property

If  $f$  is a linear transformation from  $V$  to  $W$ , then

$$f(\mathbf{0}) = \mathbf{0},$$

$$f(\alpha \mathbf{x}) = \alpha f(\mathbf{x}), \forall \alpha \in \mathbb{R}.$$

$$f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y}), \forall \mathbf{x}, \mathbf{y} \in V.$$

# Typical Example: Matrix times vector

## Example:

Suppose  $A \in \mathbb{R}^{m \times n}$  is a given real matrix.

$f(\mathbf{x}) = A\mathbf{x}$  is a **linear transformation** from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ .

Verify:

$$f(\alpha \vec{x}) = \underset{m \times n}{A} \left( \underset{1 \times 1}{\alpha} \underset{n \times 1}{\vec{x}} \right) = \alpha \left( \underset{1 \times 1}{A} \underset{m \times n}{\vec{x}} \right) = \alpha f(\vec{x})$$

Let  $A = \vec{a}^T$

$$f(\vec{x} + \vec{y}) = A(\vec{x} + \vec{y}) = \underset{\substack{\downarrow \\ \text{distributive rule}}}{A\vec{x} + A\vec{y}} = f(\vec{x}) + f(\vec{y})$$

## Example (linear function):

Suppose  $\mathbf{a} \in \mathbb{R}^n$  is a given real vector,  $\mathbf{x} \in \mathbb{R}^n$ .

$f(\mathbf{x}) = \langle \mathbf{x}, \mathbf{a} \rangle$  a linear transformation from  $\mathbb{R}^n$  to  $\mathbb{R}$ . We call it a **linear function**.

$$= \vec{a}^T \vec{x} = a_1 x_1 + a_2 x_2 + \dots + a_n x_n.$$

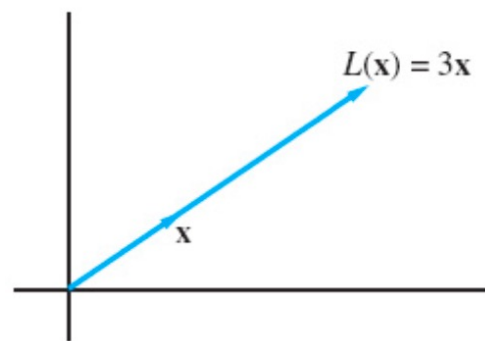
# Examples of Linear Transformations

**Example** Let  $L : \mathbb{R}^2 \rightarrow \mathbb{R}$  with  $L(x, y) = x - y$

Check linearity by Def: or simpler: write as  $A\vec{v}$

$$L(x, y) = \underbrace{x - y}_{1 \times 1} = \underbrace{[1, -1]}_{1 \times 2} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{2 \times 1}$$

**Example**  $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ . Stretching or shrinking:  $L(x, y) = (\alpha x, \alpha y)$  ( $\alpha > 0$ )



$$L(x, y) = \begin{bmatrix} \alpha x \\ \alpha y \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

# LT or not?

Example 1:  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined as  $f(x) = 2x+1$  is not a linear transformation because  $f(0) \neq 0$ .

$f(\mathbf{x}) = A\mathbf{x} + b$  is called affine transformation.

## Example 2

Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be a mapping defined as:

$$\underline{T([x, y, z]^T)} = \underline{[3x + 2y, 0, 2x + y - 1]^T}$$

$T$  is not a linear transformation since

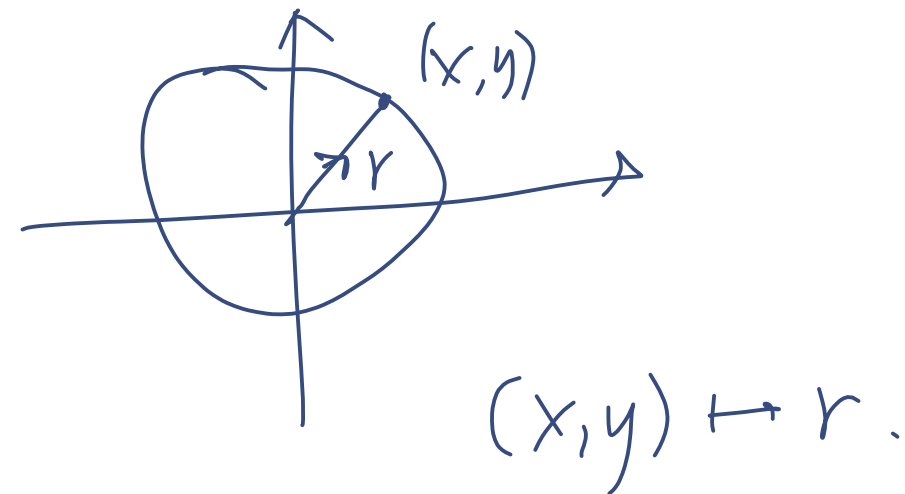
$$T(0, 0, 0) = [0, 0, -1]^T \neq \vec{0}.$$

# Non-Example of Linear Transformations

## Question

Let  $L : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a mapping defined as:

$$L(x, y) = \sqrt{x^2 + y^2}$$



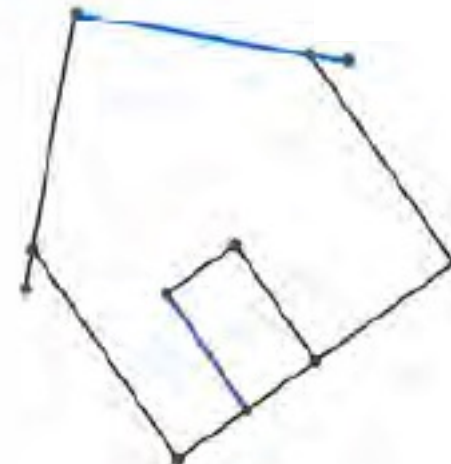
Is this  $L$  a linear transformation?

$L$  is not a linear transformation since

$$L(-(x, y)) = \underline{L(-x, -y)}$$

$$= \sqrt{(-x)^2 + (-y)^2} = \sqrt{\underbrace{x^2 + y^2}_{>0}} \neq \underbrace{-L(x, y)}_{<0}$$

# Rotating Shapes



Rotating is a “transformation”.

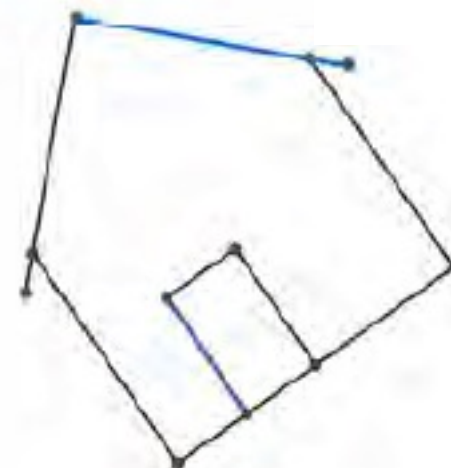
**Question 1:** How do I know it’s a **linear** transformation?

**Question 2:** How do I **derive** its expression?

# Part IV Checking Linear Transformation

15 mins

# Description of “Rotation”

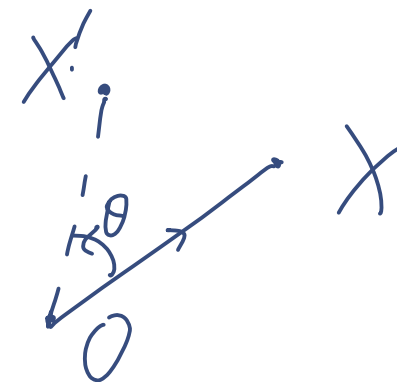


Rotation is a “transformation”.

## Precise description of rotation:

“Rotating an image around point  $O$  by degree  $\theta$ ” means:

For any point  $X$ , the new point  $X'$  satisfies  $\angle(OX, OX') = \theta$ .



**Question 1:** How do I know it's a **linear** transformation?

# Q1 How to Check Linear Transformation

**Question 1:** Is it a **linear** transformation?

**Check:** It satisfies the definition.



$$\underline{f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y}), \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n}$$

Fix  $\mathbf{x}, \mathbf{y}$ , consider all  $\alpha, \beta \in \mathbb{R}$ .

Fix  $\vec{x}, \vec{y}$  (for any).

$\alpha = \beta = 0$       **Origin to Origin.**

Next, we analyze the geometrical interpretation of the formula.

# Observation 1: Line

Observation 1: The following set represents a line.

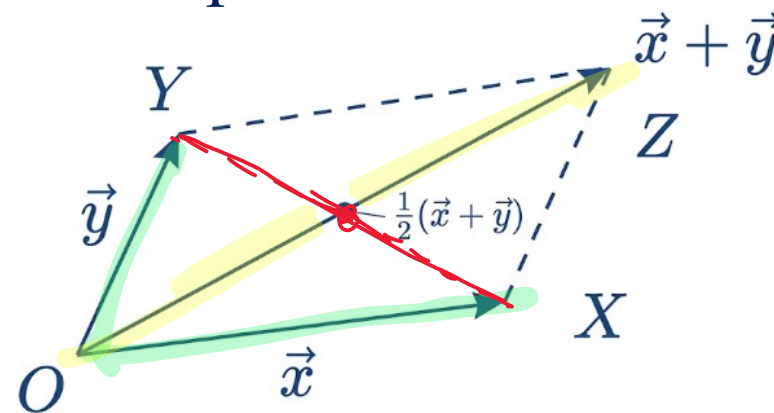
$$\Omega_1 \triangleq \{\alpha \mathbf{x} + \beta \mathbf{y} \mid \alpha + \beta = 1\}$$

one way:  
use Def of subspace.

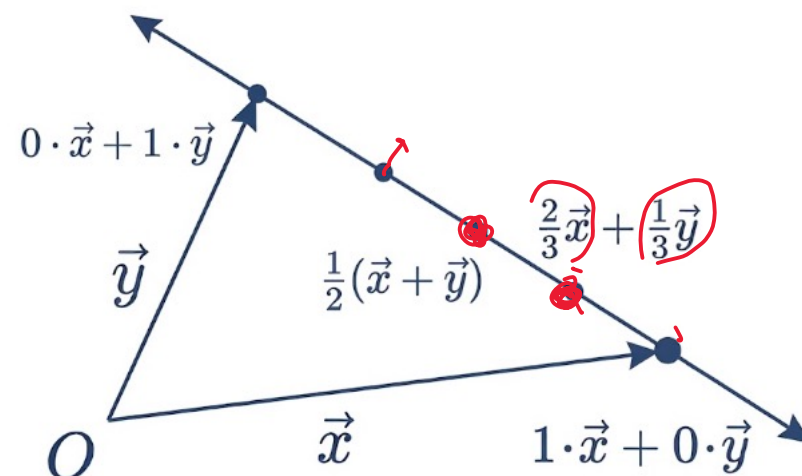
Suppose the vectors  $\mathbf{x}$ ,  $\mathbf{y}$  are  $OX$  and  $OY$  respectively.

First, we illustrate  $(\mathbf{x} + \mathbf{y})/2$  is the middle point of  $XY$ .

Middle point of  $OZ$ ,  
is the middle point of  $XY$   
(property of parallelogram)



Then, we show some points on the line  $XY$ .

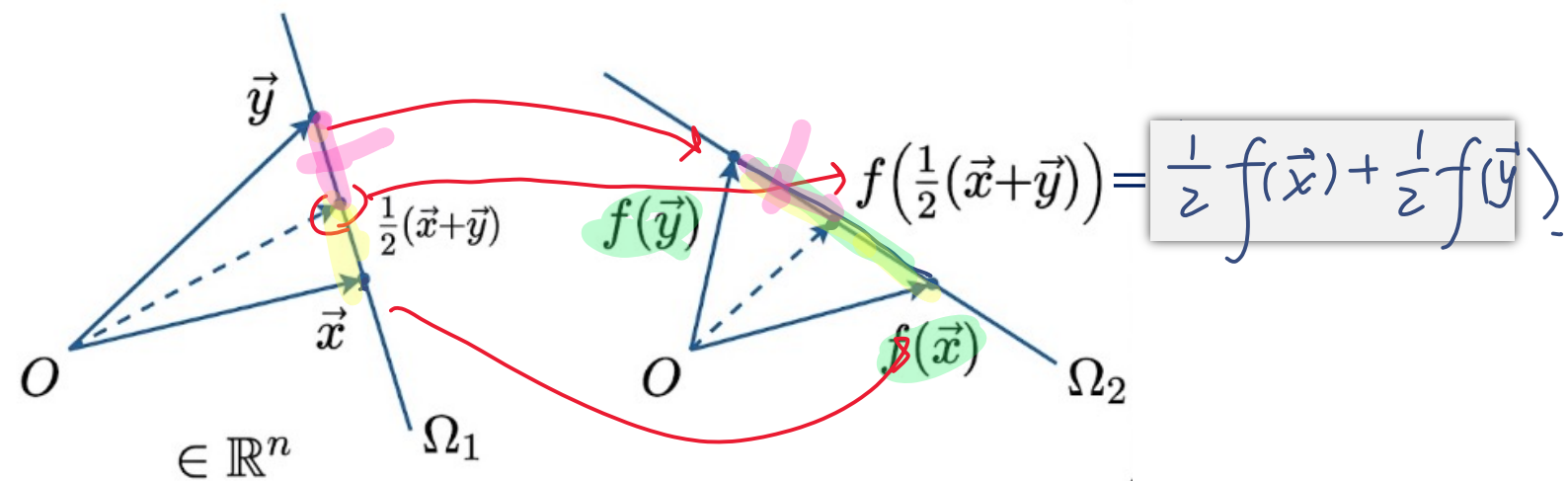


# Derivation of “Lines to Lines”

**Observation 2:** The following two sets represent two lines.

$$\Omega_1 \triangleq \{\alpha \vec{x} + \beta \vec{y} \mid \alpha + \beta = 1\}$$

$$\Omega_2 \triangleq \{\alpha f(\vec{x}) + \beta f(\vec{y}) \mid \alpha + \beta = 1\}$$



**Observation 3:**  $f$  maps one line to another; maps middle pt to middle pt.

By definition of linear transformation:  $f(\Omega_1) = \Omega_2$ , i. e., line mapped to line

$$\{f(\alpha \vec{x} + \beta \vec{y}) \mid \alpha + \beta = 1\} = \{\alpha f(\vec{x}) + \beta f(\vec{y}) \mid \alpha + \beta = 1\}$$

Let  $\alpha = \beta = \frac{1}{2}$ :  $f(\frac{1}{2}\vec{x} + \frac{1}{2}\vec{y}) = \frac{1}{2}f(\vec{x}) + \frac{1}{2}f(\vec{y})$ , or middle pt  $\rightarrow$  middle pt.

# Checking Linear Transformation: Geometry

**Question 1:** Is a mapping  $T$  a **linear** transformation?

$$T(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha T(\mathbf{x}) + \beta T(\mathbf{y}), \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Fix  $\mathbf{x}, \mathbf{y}$ , consider all  $\alpha, \beta \in \mathbb{R}$ .

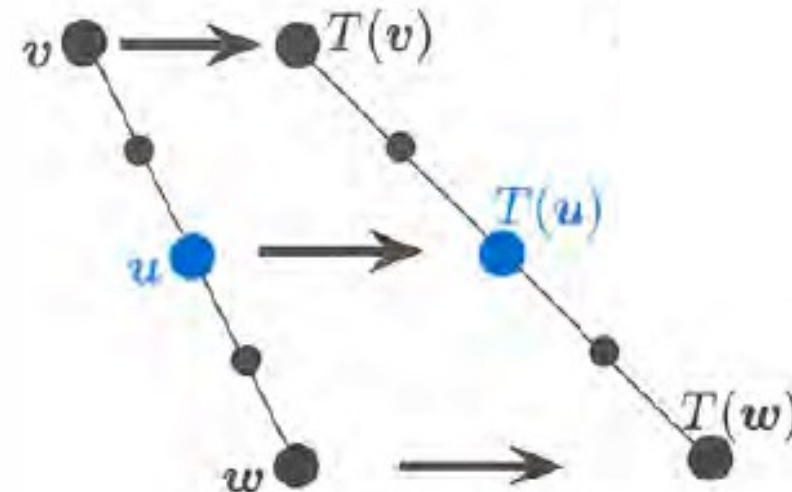
$$\alpha + \beta = 1$$

**Rule 1:** Lines to Lines.

**Rule 2:** Equally spaced points to equally spaced points.

$$\alpha = \beta = 0$$

**Rule 3:** Origin to Origin.



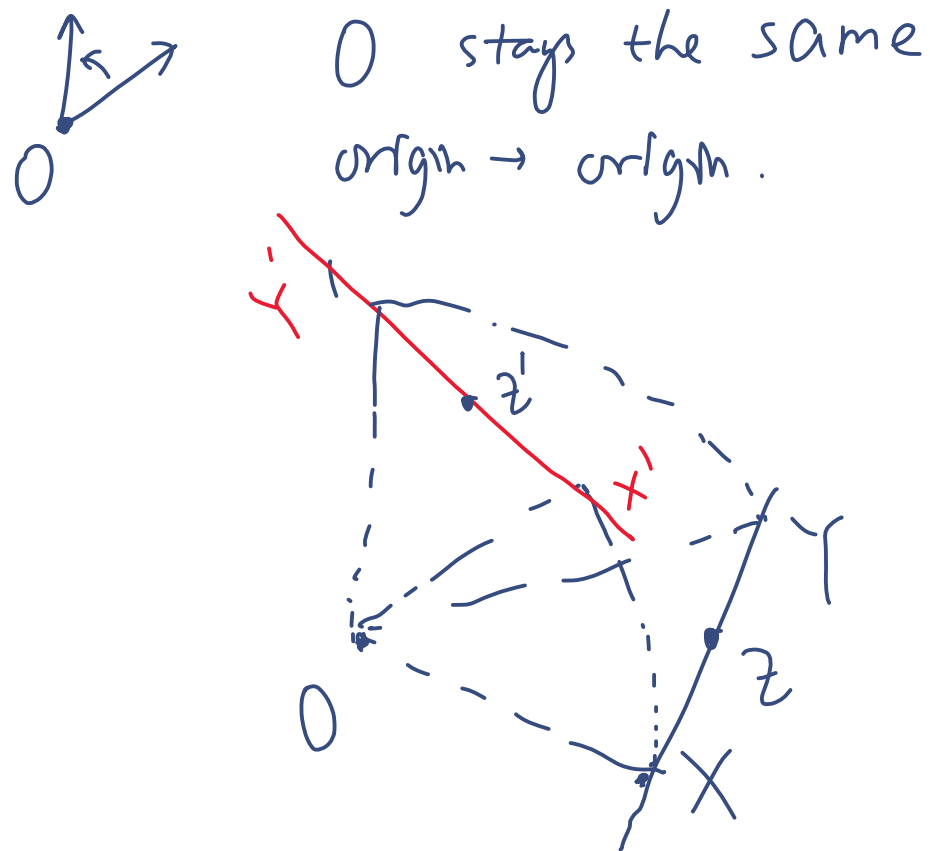
$$\Omega_1 \triangleq \{\alpha \mathbf{w} + \beta \mathbf{v} \mid \alpha + \beta = 1\} \quad \Omega_2 \triangleq \{\alpha T(\mathbf{w}) + \beta T(\mathbf{v}) \mid \alpha + \beta = 1\}$$

# Check Rotation

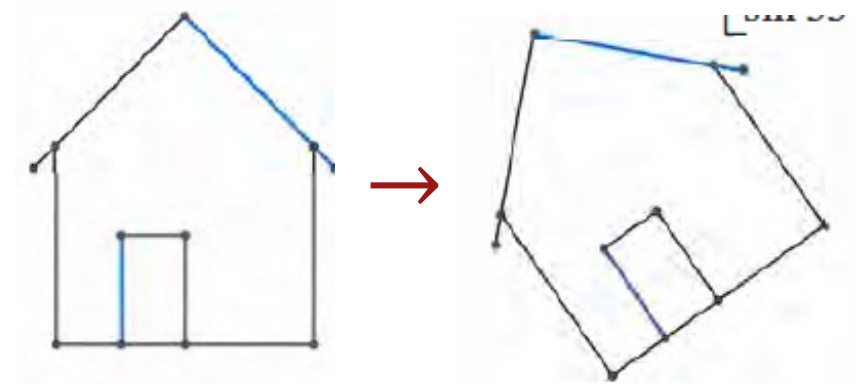
“Rotating an image around point  $O$  by degree  $\theta$ ” means:  
For any point  $X$ , the new point  $X'$  satisfies  $\angle(OX, OX') = \theta$ .

**Question 1:** Is it a **linear** transformation?

**Check:** It satisfies Rule 1, 2, 3.



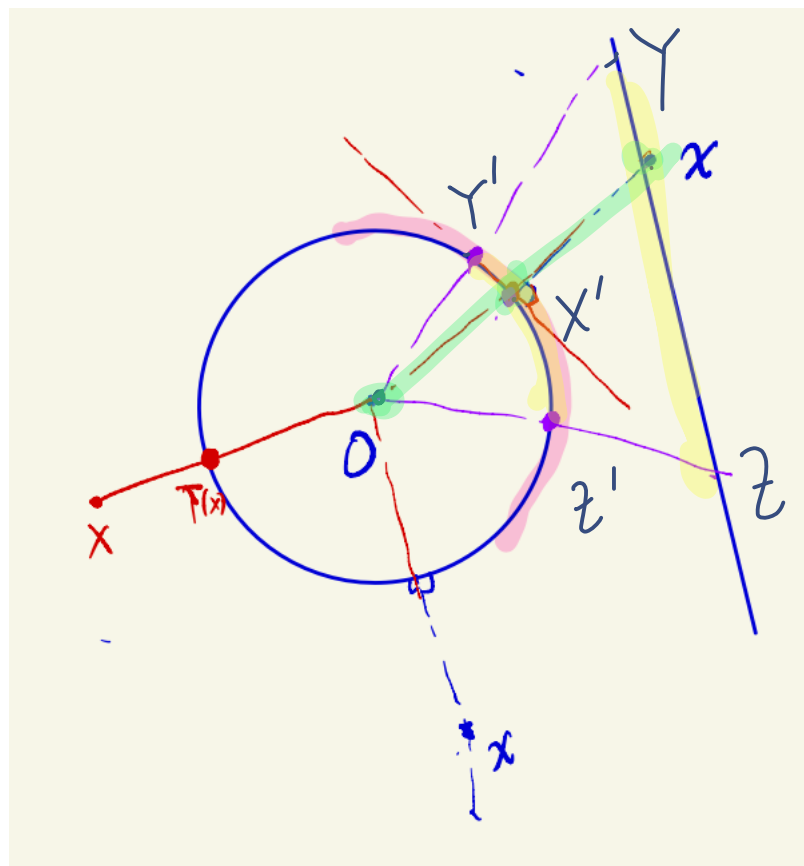
line  $XY$   
 $\rightarrow X'Y'$ , a line  
middle pt  $Z$  of  $XY$   
 $\rightarrow Z'$ , a middle pt of  $X'Y'$ .



## Eg: Projection onto Circles

Given a circle  $C$  centered at the origin  $O$ . For any point  $X \in \mathbb{R}^2$ , connect  $O$  to  $X$ , and let  $X'$  be the intersection of line  $OX$  and circle  $C$ . Define  $T(X) = X'$ .

**Question:** Is  $T$  a linear transformation?



No.  
line  $\rightarrow$  arc  
not L.T.

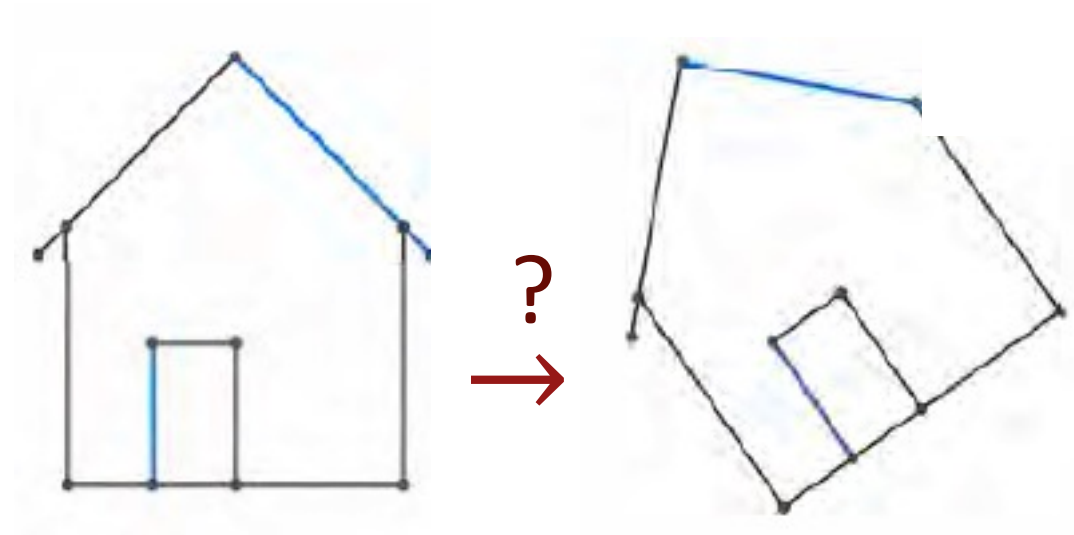
# Part V Matrix of Linear Transformation

Strang's book v5, Sec 8.2

Session "Construction of the matrix"

25 mins

## Q2: Math Form of Rotation



**Q2:** How to **derive** its expression?

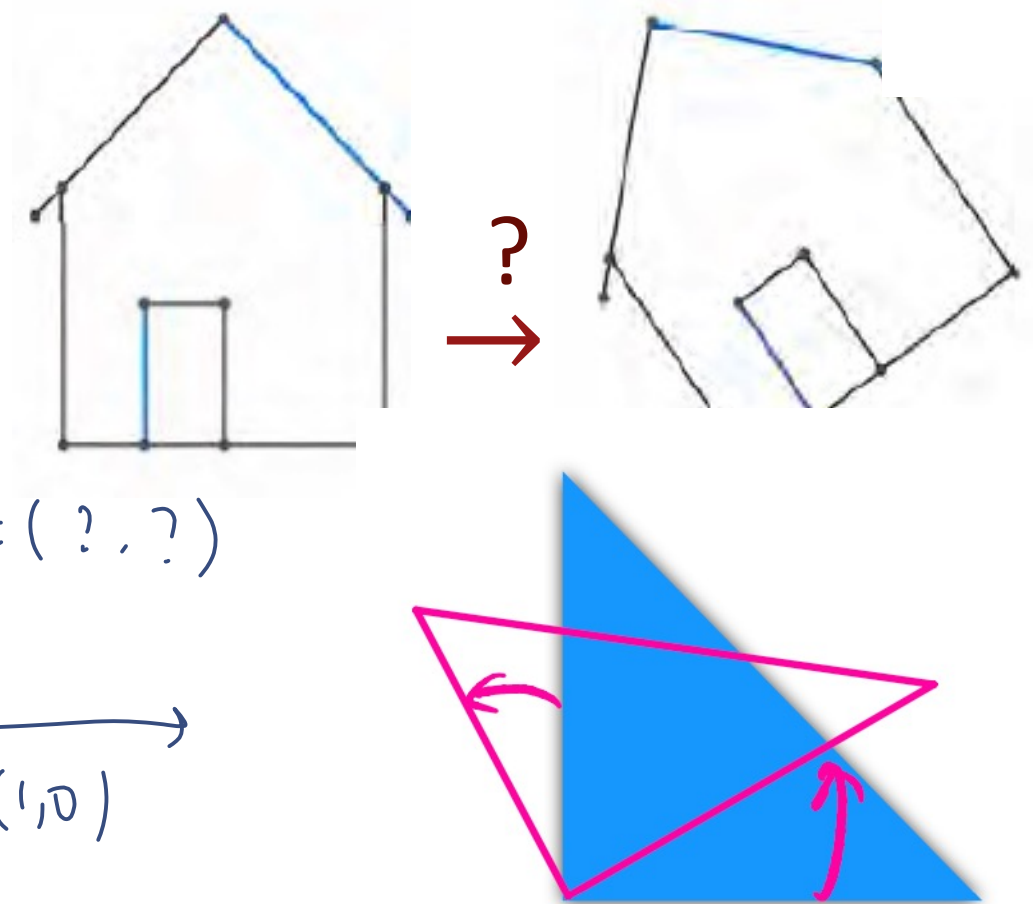
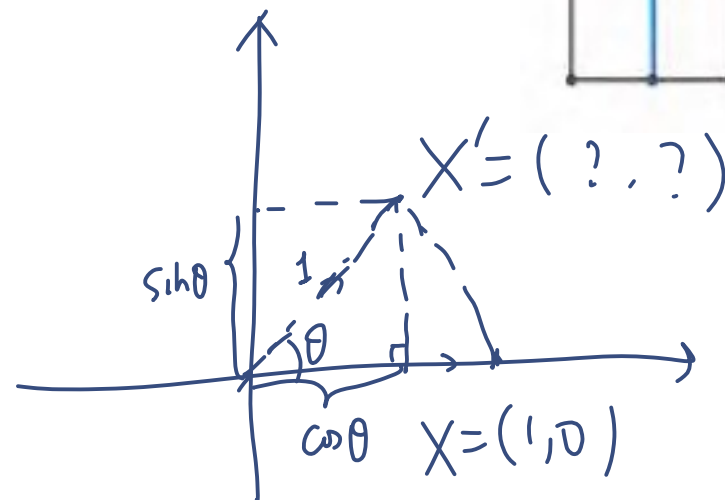
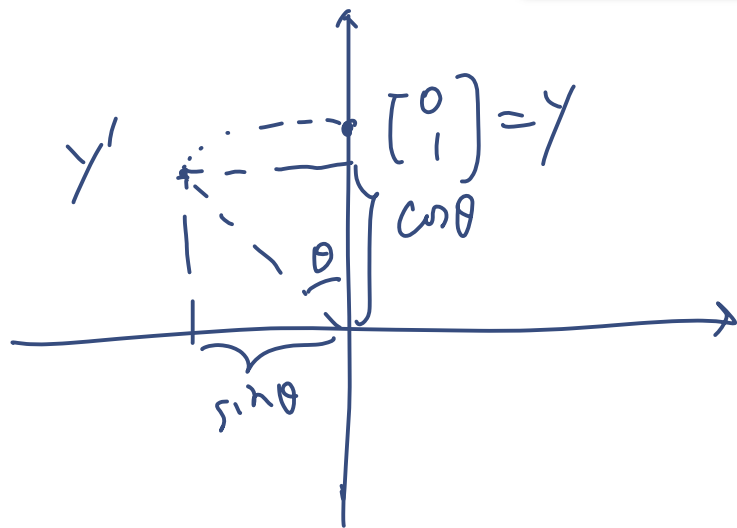
## Q2: Math Form of Rotation

Q2: How to **derive** its expression?

**From basis!**

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto T(\mathbf{e}_1) = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto T(\mathbf{e}_2) = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$



## Q2: Math Form of Rotation

Q2: How to **derive** its expression?

**From basis!**

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto T(\mathbf{e}_1) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto T(\mathbf{e}_2) = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto T(\mathbf{x})$$
$$= x_1 \vec{e}_1 + x_2 \vec{e}_2$$

$$T(\mathbf{x}) = T(x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2)$$

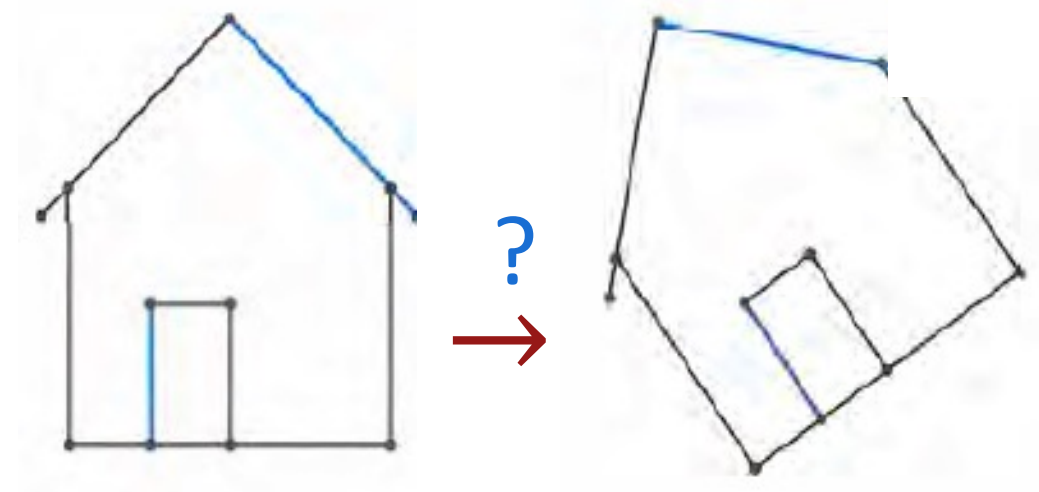
(def of linearity)

$$= x_1 T(\vec{e}_1) + x_2 T(\vec{e}_2)$$
$$= x_1 \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} + x_2 \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A\mathbf{x}$$

where

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = [T(\vec{e}_1) \quad T(\vec{e}_2)]$$



# Matrix of Linear Transformation

## Theorem 20.1 *transformation*

For any linear ~~mapping~~ *transformation*  $f$  from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ , there exists a matrix  $A$  such that  $f(x) = Ax$ .

In other words, if  $f$  satisfies

$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y}), \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n,$   
then  $f(x) = Ax$  for some  $A$ .

# Key Steps of Proof

~~Claim~~ Def

If  $f$  is a linear transformation from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ , then

$$f\left(\sum_{i=1}^k \alpha_i \mathbf{v}_i\right) = \sum_{i=1}^k \alpha_i f(\mathbf{v}_i), \forall \alpha_i \in \mathbb{R}, \forall \mathbf{v}_i \in \mathbb{R}^n, i = 1, \dots, k.$$

## Proof of Theorem 20.1:

Pick the standard basis  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  of  $\mathbb{R}^n$ .

$$f(\mathbf{x}) = f\left(\sum_i \chi_i \vec{e}_i\right) = \sum_i \chi_i f(\vec{e}_i) = A\mathbf{x}.$$

$$\text{where } A = \begin{bmatrix} f(\vec{e}_1) & \dots & f(\vec{e}_n) \end{bmatrix} \in \mathbb{R}^{m \times n}$$

# Computing <sup>matrix of</sup> Linear Transformation

## Algorithm 20.1

**Step 1:** Pick the standard basis  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  of  $\mathbb{R}^n$ .

**Step 2:** Compute  $f(\mathbf{e}_i) \in \mathbb{R}^m, \forall i$ .

**Step 3:** Form a matrix  $A = [f(\mathbf{e}_1), f(\mathbf{e}_2), \dots, f(\mathbf{e}_n)] \in \mathbb{R}^{m \times n}$

**Conclusion:** For any  $\mathbf{x} \in \mathbb{R}^n$ , we have

$$f(\mathbf{x}) = \sum_i x_i f(\mathbf{e}_i) = A\mathbf{x}.$$

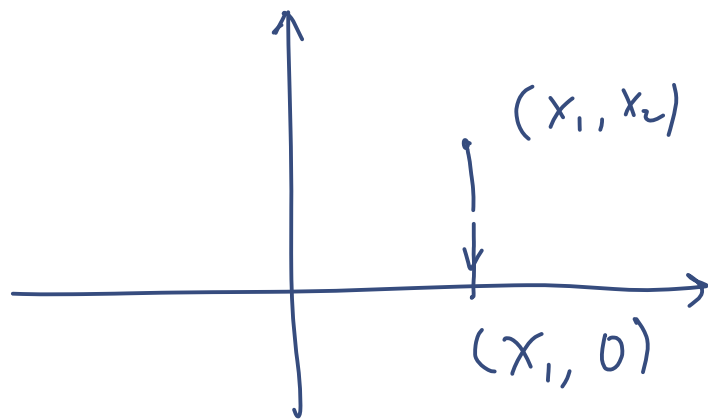
$A$  is the matrix of the linear transformation  $f$ .

## Exercise: Checking Linear Transformation & Finding Matrix

Exercise: Is the following mapping  $T(\cdot)$  a linear transformation?

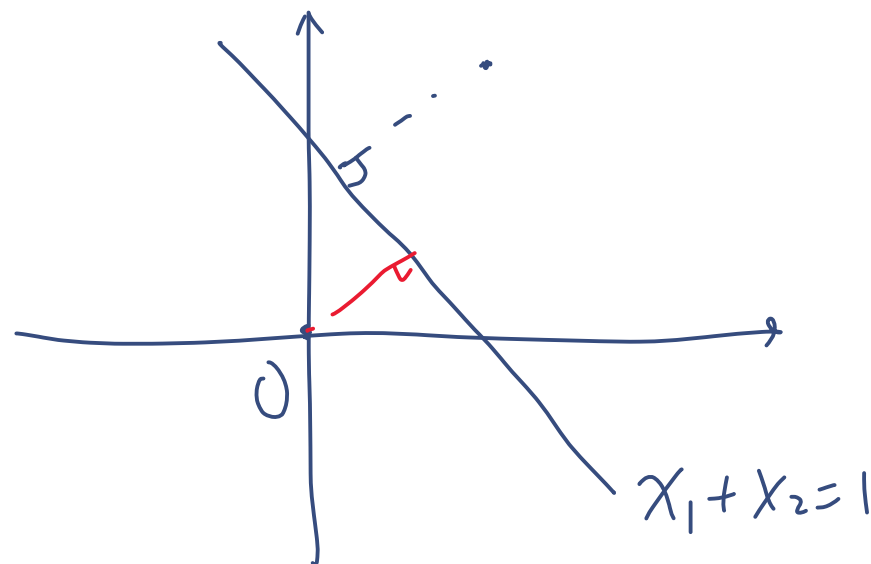
If so, find a matrix  $A$  such that  $T(x) = Ax$ .

**E1** Suppose  $T(x)$  is the **projection** of  $(x_1, x_2)$  onto  $x_1$ -axis.



$$T(x_1, x_2) = \underbrace{(x_1, 0)}_{2 \times 1} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{2 \times 1}$$

**E2** Suppose  $T(y)$  is the **projection** of  $(y_1, y_2)$  onto  $\{(x_1, x_2) \mid x_1 + x_2 = 1\}$ .



No.

$$T(0,0) \neq (0,0)$$

## Comments on Two Problem Types

### Type 1:

We know it is a linear transformation.

Find a matrix  $A$  such that  $T(x) = Ax$ .

#### Solution method:

T1M1: write its expression, then write it as  $Ax$  directly;

T1M2: use Algorithm 20.1.

### Type 2:

The problem asks whether it is a linear transformation, and find a matrix  $A$  such that  $T(x) = Ax$ .

#### Solution methods:

T2M1 (not that common though): test whether it is linear transformation first; then use Algorithm 20.1.

→ w + a + bn

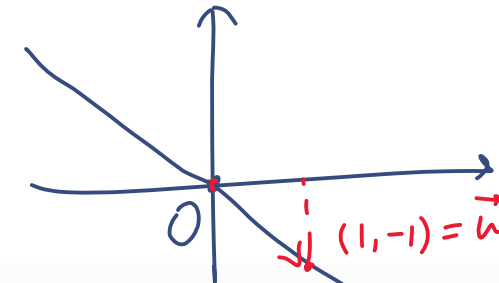
T2M2: derive its expression, then write it as  $Ax$  directly (which automatically implies it is linear transformation).

## Exercise: Finding Matrix

**E3** Suppose  $T(y)$  is the projection of  $(y_1, y_2)$  onto  $\{(x_1, x_2) \mid x_1 + x_2 = 0\}$ . We know it is a linear transformation. Find a matrix  $A$  such that  $T(x) = Ax$ .

### Method 1:

**Step 1:** Find an orthonormal basis of  $W$ .



The subspace  $W$  is the line  $x_1 + x_2 = 0$ , which is spanned by the vector  $w = (1, -1)$ .

$u = \frac{w}{\|w\|} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$  forms an orthonormal basis of  $W$ .

**Step 2:** Compute the projection of  $e_1, e_2$ .

$$\text{proj}_W(v) = (u^\top v)u.$$

$$u^\top e_1 = \frac{1}{\sqrt{2}}(1, -1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}. \quad \Rightarrow \quad T(e_1) = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}.$$

$$u^\top e_2 = \frac{1}{\sqrt{2}}(1, -1) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{\sqrt{2}}. \quad \Rightarrow \quad T(e_2) = -\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \end{bmatrix}.$$

**Step 3:** Form the matrix  $A = [T(e_1), T(e_2)]$ .

$$A = [T(e_1) \quad T(e_2)] = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}.$$

**Conclusion:** The matrix  $A$  is  $\begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$ .

## Exercise: Checking Linear Transformation & Finding Matrix

**E3-b** Suppose  $T(y)$  is the projection of  $(y_1, y_2)$  onto  $\{(x_1, x_2) \mid x_1 + x_2 = 0\}$ .

Is the mapping  $T(\cdot)$  a linear transformation?

If so, find a matrix  $A$  such that  $T(x) = Ax$ .

**Step 1:** Find an orthonormal basis of  $W$ .

The subspace  $W$  is the line  $x_1 + x_2 = 0$ , which is spanned by the vector  $w = (1, -1)$ .

$u_1 = \frac{w}{\|w\|} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$  forms an orthonormal basis of  $W$ .

**Step 2:** Write the formula for the expression of the projection.

$$T(y) = \text{proj}_W(y) = (u_1^\top y) u_1$$

$$\Rightarrow T(y) = \frac{y_1 - y_2}{\sqrt{2}} \cdot \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = \left(\frac{y_1 - y_2}{2}, -\frac{y_1 - y_2}{2}\right)$$

**Step 3:** Represent  $T$  as a matrix.

$$T \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \frac{y_1 - y_2}{2} \\ -\frac{y_1 - y_2}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Conclusion:  $T$  is a linear transformation; the matrix  $A$  is

$$\begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

# Relation and Difference?

Linear transformation

Linear system of equations

Math form

$$\mathbf{y} = A\mathbf{x}$$

$$A\mathbf{x} = \mathbf{b}$$

Applications

Rotating images

Compute price;  
Compute references...

Known

$\mathbf{x}$ ; "description" of  $\mathbf{y}$

$A, \mathbf{b}$

Ultimate goal

Compute  $\mathbf{y}$

Compute  $\mathbf{x}$

Intermediate

obtain  $A$



# Summary Today (Write Your Own)

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**One sentence summary:**

**Detailed summary:**

# Summary Today (Instructor's version)

---

## One sentence summary:

We learned general linear transformation and how to “express” it.

## Detailed summary:

### Properties of Determinants:

- $\det(AB) = \det(A) \det(B)$
- $\det(A) \neq 0$  iff  $A$  is invertible

### Check Linear transformation (Euclidean space)

Origin to origin; line to line; middle pt to middle pt.

### Deriving Expression of Linear transformation (Euclidean space)

Find basis;

Find the transformed vectors.

Put in columns of a matrix.

—> Matrix of linear transformation.

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